Temperature-sensitive products, such as fresh produce, dairy, and frozen foods, require constant monitoring to prevent spoilage. Transportation networks move products across varying climate conditions, with refrigerated trucks facing compressor failures, door seal degradation, and inadequate airflow that create hot spots and cold spots within individual trailers.

Store-level operations present their own challenges. Inventory accuracy has a direct impact on both customer satisfaction and capital efficiency; however, physical inventory counts are labour-intensive and error-prone. Stockouts drive customers to competitors while overstocking ties up capital and leads to markdowns or waste. Theft and shrinkage drain profitability, with traditional loss prevention focused on detection rather than prevention. Customer behaviour analytics remain shallow, based on point-of-sale data that reveals what people bought but not what they considered, what they couldn't find, or how they navigated the store.

Cloud-centric approaches to these challenges face fundamental limitations. Sending every RFID read event, every computer vision frame, and every sensor measurement to the cloud creates untenable bandwidth costs and latency that prevent real-time intervention. A single distribution centre with a thousand cameras generating standard-definition video at 15 frames per second produces over 200 gigabytes per hour of raw data. Multiply across hundreds of locations and the costs become prohibitive.

Edge AI architecture: The intelligent retail supply chain deploys edge computing infrastructure at every level. Distribution centres receive powerful edge gateway servers running K3s clusters, hosting computer vision models for visual inspection, TensorFlow Lite inference engines for quality prediction, and EdgeX Foundry for device integration. Refrigerated transport vehicles carry ruggedized edge computers with cellular connectivity, running anomaly detection models that predict equipment failures hours before breakdown. Individual store locations deploy edge gateways coordinating networks of smart shelves, computer vision cameras, and environmental sensors.

At the device level, smart shelves equipped with weight sensors and RFID readers continuously track inventory levels and product movement patterns. Computer vision cameras monitor produce quality, detecting indicators of bruising, discolouration, or spoilage. Environmental sensors throughout the cold chain track not just temperature and humidity but also vibration, light exposure, and gas composition, building a rich context for predictive quality models. Each device runs lightweight AI models that perform local inference, publishing only actionable insights and anomalies, rather than raw sensor streams.

The communication layer implements MQTT for reliable, bandwidth-efficient telemetry with quality of service guarantees, ensuring that critical alerts are never lost. Devices publish to hierarchical topics, enabling flexible subscription patterns. Store-level gateways aggregate local data and forward summary statistics and exceptions to regional and enterprise tiers. The protocol layer implements store-and-forward logic, buffering data during network outages and intelligently prioritising transmission when connectivity is restored, ensuring that high-priority alerts always reach their destination, even under adverse conditions.

EdgeX Foundry at the gateway tier integrates diverse devices through protocol-specific device services, normalising data into consistent formats and providing unified access for higher-level services. The rules engine implements local business logic, automatically triggering replenishment orders when inventory falls below safety stock levels, alerting staff when temperature excursions threaten product quality, and coordinating responses across multiple systems. Application services filter and transform data for export to enterprise systems, implementing rate limiting, privacy filtering, and protocol translation.

The AI inference layer runs sophisticated models trained on historical data and continuously refined through federated learning. Quality prediction models analyse and produce images, predicting remaining shelf life and optimal markdown timing. Demand forecasting models combine historical sales patterns, local events, weather forecasts, and real-time foot traffic to optimise inventory positioning. Anomaly detection models identify unusual patterns suggesting equipment problems, security concerns, or process breakdowns.

The analytics pipeline aggregates insights across locations, identifying patterns invisible at individual sites. Apache Spark Streaming processes event streams, correlating inventory levels across distribution networks to optimise allocation. Time-series analysis identifies systematic problems with suppliers, transportation providers, or specific SKUs.

The intelligent retail supply chain delivers transformative improvements across multiple dimensions. Inventory accuracy improves from typical levels of 85% to 98%. This directly translates to reduced stockouts, with availability improving from 92% to 98%. This captures sales previously lost to out-of-stock conditions while simultaneously reducing safety stock requirements and capital tied up in inventory.

Waste reduction is equally dramatic, particularly for perishable products. Predictive quality models enable dynamic markdown timing, selling products while they are still fresh but approaching the end of their shelf life, rather than holding them for full price until spoilage forces disposal. Smart cold chain monitoring detects equipment problems and temperature excursions